6.5 Differentiation from Axioms

WORKSHEET

The question

STEP II 1992, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/92-s2-q4

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself Rule (i) is the product rule. How could the product rule tell you anything about a constant? (Think of a constant as a product.)

STAGE 1 Constants

Ask yourself Apply rule (i) with $\mathrm{f} = \mathrm{g} = 1$. Then use rule (iv) for any other constant.

STAGE 2 Powers

Pause and try Prove by induction that $\Diamond x^n = n x^{n-1}$ for every positive integer n.

Try this next

1. **A different derivation.** Suppose $\Diamond$ satisfies rules (i), (ii) and (iv), but $\Diamond x = x$ instead of 1. Show that $\Diamond \mathrm{h}(x) = x\,\mathrm{h}'(x)$. What does $\Diamond$ do to x^n?

2. **The quotient rule.** Extend $\Diamond$ to rational functions f/g, assuming rule (i) still holds. By applying rule (i) to $\mathrm{g} \cdot \frac{\mathrm{f}}{\mathrm{g}} = \mathrm{f}$, derive the quotient rule.

3. **Repeated roots.** Show that if $(x-a)^2$ divides a polynomial f, then $x-a$ divides f'. Use this to show that $x^n - 1$ has no repeated roots.

4. **Dual numbers.** Use dual numbers to compute $f(2)$ and $f'(2)$ for $f(x) = (x^2+1)^3$, and check with the chain rule.

Further reading

- **Peter Cameron, *Introduction to Algebra*** (Oxford University Press). Rings, polynomials and fields at first-year level, with the formal derivative.
- **Andreas Griewank & Andrea Walther, *Evaluating Derivatives*** (SIAM). The standard reference on automatic differentiation. Advanced.
- **Michael Nielsen, *Neural Networks and Deep Learning*** (free online). Chapter 2 explains backpropagation clearly.

The full guided solution, and what the question leads to at university, is in Section 6.5:
https://beyondstep.keithmcnulty.org/ch06/derivations.html